



User's Query: What's the fastest lap time ever in a race for Lewis Hamilton?

CHESSE MULTI-AGENT SYSTEM

EXTRACT KEYWORDS:

- "FASTEST LAP TIME", "LEWIS HAMILTON"

RETRIEVE ENTITY: TABLE DRIVERS HAS COLUMNS

- SURNAME: ["HAMILTON"]
- FORENAME: ["LEWIS"]

RETRIEVE CONTEXT: FASTESTLAPTIME

- COLUMN NAME: "FASTEST LAP TIME"
- COLUMN DESCRIPTION: "FASTEST LAP TIME"
- VALUE DESCRIPTION: 'FASTESTLAPTIME' LEADS TO HIGHER RANK



INFORMATION RETRIEVER



CANDIDATE GENERATOR

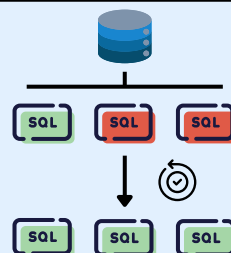


GENERATE CANDIDATE QUERY:

```
SELECT MIN(T2.FASTESTLAPTIME)
FROM DRIVERS AS T1 INNER JOIN RESULTS
AS T2 ON T1.DRIVERID = T2.DRIVERID
WHERE T1.FORENAME = 'LEWIS' AND
T1.SURNAME = 'HAMILTON'
```

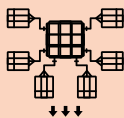
REVISE:

```
SELECT MIN(T2.FASTESTLAPTIME)
FROM DRIVERS AS T1 INNER JOIN RESULTS
AS T2 ON T1.DRIVERID = T2.DRIVERID
WHERE T1.FORENAME = 'LEWIS' AND
T1.SURNAME = 'HAMILTON'
```



FILTER_COLUMN:

CONSTRUCTORS: [NAME, CONSTRUCTORID]
DRIVERS: [DRIVERREF, FORENAME, SURNAME, DRIVERID]
SEASONS: [YEAR]
RACES: [TIME, YEAR]

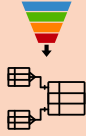


SELECT TABLES:

SELECTED TABLES : DRIVERS, RESULTS

SELECT COLUMNS:

DRIVERS: [DRIVERID , FORENAME, SURNAME]
RESULTS: [DRIVERID, FASTESTLAPTIME]



SCHEMA SELECTOR



UNIT TESTER



GENERATE UNIT TEST:

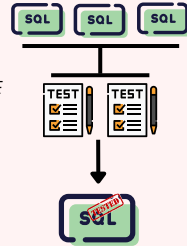
- UNIT TEST 1:** THE ANSWER SQL QUERY SHOULD SELECT THE FASTESTLAPTIME COLUMN.
- UNIT TEST 2:** THE ANSWER SQL QUERY SHOULD JOIN THE 'RESULTS' TABLE AND THE 'DRIVERS' TABLE.

EVALUATE:

SCORES:

QUERY 1: **2**, QUERY 2: **4**, QUERY 3: **7**, QUERY 4: **1**

- Query 3 with the highest score is the final SQL query.



```
SELECT T2.FASTESTLAPTIME
FROM DRIVERS AS T1 INNER JOIN RESULTS AS T2 ON T2.DRIVERID = T1.DRIVERID
WHERE T1.FORENAME = 'LEWIS' AND T1.SURNAME = 'HAMILTON' AND T2.FASTESTLAPTIME IS NOT NULL
ORDER BY T2.FASTESTLAPTIME ASC LIMIT 1
```